```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

df = pd.read_csv("/content/Zomato Dataset.csv")
df.head()
```

	ID	Delivery_person_ID	Delivery_person_Age	Delivery_person_Ratings	Restaurant_latitude	Restaurant_longitude	Deliver
0	0xcdcd	DEHRES17DEL01	36.0	4.2	30.327968	78.046106	
1	0xd987	KOCRES16DEL01	21.0	4.7	10.003064	76.307589	
2	0x2784	PUNERES13DEL03	23.0	4.7	18.562450	73.916619	
3	0xc8b6	LUDHRES15DEL02	34.0	4.3	30.899584	75.809346	
4	0xdb64	KNPRES14DEL02	24.0	4.7	26.463504	80.372929	

Next steps: Generate code with df New interactive sheet

```
print("Shape:", df.shape)
print("\nColumns:\n", df.columns.tolist())
print("\nMissing values:\n", df.isnull().sum())
df.info()
```

```
Shape: (45584, 20)

Columns:
 ['ID', 'Delivery_person_ID', 'Delivery_person_Age', 'Delivery_person_Ratings', 'Restaurant_latitude', 'Restaurant_longitude',

Missing values:
 ID                                0
Delivery_person_ID                0
Delivery_person_Age            1854
Delivery_person_Ratings        1908
Restaurant_latitude               0
Restaurant_longitude              0
Delivery_location_latitude        0
Delivery_location_longitude       0
Order_Date                        0
Time_Orderd                    1731
Time_Order_picked                 0
Weather_conditions              616
Road_traffic_density            601
Vehicle_condition                 0
Type_of_order                     0
Type_of_vehicle                   0
multiple_deliveries             993
Festival                        228
City                           1200
Time_taken (min)                  0
dtype: int64
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 45584 entries, 0 to 45583
Data columns (total 20 columns):
 #   Column                       Non-Null Count  Dtype
---  ------                       --------------  -----
 0   ID                           45584 non-null  object
 1   Delivery_person_ID           45584 non-null  object
 2   Delivery_person_Age          43730 non-null  float64
 3   Delivery_person_Ratings      43676 non-null  float64
 4   Restaurant_latitude          45584 non-null  float64
 5   Restaurant_longitude         45584 non-null  float64
 6   Delivery_location_latitude   45584 non-null  float64
 7   Delivery_location_longitude  45584 non-null  float64
 8   Order_Date                   45584 non-null  object
 9   Time_Orderd                  43853 non-null  object
 10  Time_Order_picked            45584 non-null  object
 11  Weather_conditions           44968 non-null  object
 12  Road_traffic_density         44983 non-null  object
 13  Vehicle_condition            45584 non-null  int64
 14  Type_of_order                45584 non-null  object
 15  Type_of_vehicle              45584 non-null  object
 16  multiple_deliveries          44591 non-null  float64
 17  Festival                     45356 non-null  object
 18  City                         44384 non-null  object
 19  Time_taken (min)             45584 non-null  int64
dtypes: float64(7), int64(2), object(11)
memory usage: 7.0+ MB
```

```
df.columns = df.columns.str.strip().str.replace(" ", "_").str.replace("-", "_")
print(df.columns.tolist())
```

```
['ID', 'Delivery_person_ID', 'Delivery_person_Age', 'Delivery_person_Ratings', 'Restaurant_latitude', 'Restaurant_longitude',
```

```
if "Time_taken(min)" in df.columns:
    df.rename(columns={"Time_taken(min)": "Time_taken_min"}, inplace=True)
elif "Time_taken_(min)" in df.columns:
    df.rename(columns={"Time_taken_(min)": "Time_taken_min"}, inplace=True)
elif "Time_taken_(min)" not in df.columns and "Time_taken_min" not in df.columns:
    for col in df.columns:
        if "Time_taken" in col:
            df.rename(columns={col: "Time_taken_min"}, inplace=True)

print(df.columns.tolist())
```

```
['ID', 'Delivery_person_ID', 'Delivery_person_Age', 'Delivery_person_Ratings', 'Restaurant_latitude', 'Restaurant_longitude',
```

```
numeric_cols = [
    "Delivery_person_Age",
    "Delivery_person_Ratings",
    "Restaurant_latitude",
    "Restaurant_longitude",
    "Delivery_location_latitude",
    "Delivery_location_longitude",
    "Vehicle_condition",
    "multiple_deliveries",
    "Time_taken_min"
]

for col in numeric_cols:
    if col in df.columns:
        df[col] = pd.to_numeric(df[col], errors="coerce")

print(df[numeric_cols].head())
```

```
   Delivery_person_Age  Delivery_person_Ratings  Restaurant_latitude  \
0                 36.0                      4.2            30.327968
1                 21.0                      4.7            10.003064
2                 23.0                      4.7            18.562450
3                 34.0                      4.3            30.899584
4                 24.0                      4.7            26.463504

   Restaurant_longitude  Delivery_location_latitude  \
0             78.046106                   30.397968
1             76.307589                   10.043064
2             73.916619                   18.652450
3             75.809346                   30.919584
4             80.372929                   26.593504

   Delivery_location_longitude  Vehicle_condition  multiple_deliveries  \
0                    78.116106                  2                  3.0
1                    76.347589                  1                  1.0
2                    74.006619                  1                  1.0
3                    75.829346                  0                  0.0
4                    80.502929                  1                  1.0

   Time_taken_min
0              46
1              23
2              21
3              20
4              41
```

```
df["Order_Date"] = pd.to_datetime(df["Order_Date"], format="%d-%m-%Y", errors="coerce")

df["Time_Orderd"] = pd.to_datetime(df["Time_Orderd"], format="%H:%M", errors="coerce")
df["Time_Order_picked"] = pd.to_datetime(df["Time_Order_picked"], format="%H:%M", errors="coerce")
```

```
def time_diff_minutes(start, end):
    if pd.isnull(start) or pd.isnull(end):
        return np.nan
    diff = (end - start).total_seconds() / 60
    if diff < 0:
        diff += 24 * 60
    return diff

df["prep_time_min"] = df.apply(
    lambda row: time_diff_minutes(row["Time_Orderd"], row["Time_Order_picked"]), axis=1
)
```

```
df[["Time_Orderd", "Time_Order_picked", "prep_time_min"]].head()
```

	Time_Orderd	Time_Order_picked	prep_time_min
0	1900-01-01 21:55:00	1900-01-01 22:10:00	15.0
1	1900-01-01 14:55:00	1900-01-01 15:05:00	10.0
2	1900-01-01 17:30:00	1900-01-01 17:40:00	10.0
3	1900-01-01 09:20:00	1900-01-01 09:30:00	10.0
4	1900-01-01 19:50:00	1900-01-01 20:05:00	15.0

```
def haversine(lat1, lon1, lat2, lon2):
    R = 6371
    lat1, lon1, lat2, lon2 = map(np.radians, [lat1, lon1, lat2, lon2])
    dlat = lat2 - lat1
    dlon = lon2 - lon1
    a = np.sin(dlat / 2)**2 + np.cos(lat1) * np.cos(lat2) * np.sin(dlon / 2)**2
    c = 2 * np.arcsin(np.sqrt(a))
    return R * c

df["distance_km"] = haversine(
    df["Restaurant_latitude"],
    df["Restaurant_longitude"],
    df["Delivery_location_latitude"],
    df["Delivery_location_longitude"]
)

df[["Restaurant_latitude", "Restaurant_longitude",
    "Delivery_location_latitude", "Delivery_location_longitude",
    "distance_km"]].head()
```

	Restaurant_latitude	Restaurant_longitude	Delivery_location_latitude	Delivery_location_longitude	distance_km
0	30.327968	78.046106	30.397968	78.116106	10.280582
1	10.003064	76.307589	10.043064	76.347589	6.242319
2	18.562450	73.916619	18.652450	74.006619	13.787860
3	30.899584	75.809346	30.919584	75.829346	2.930258
4	26.463504	80.372929	26.593504	80.502929	19.396618

```
df = df[
    (df["Time_taken_min"].notnull()) &
    (df["distance_km"].notnull()) &
    (df["distance_km"] > 0)
]

df = df[df["Time_taken_min"] > 0]

print("Cleaned shape:", df.shape)
```

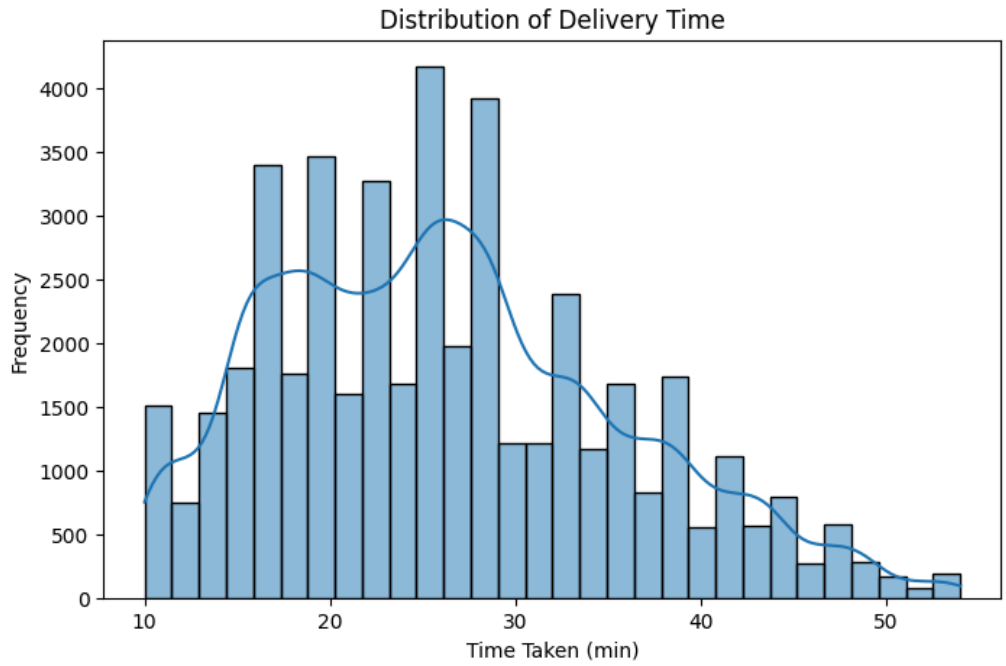
```
Cleaned shape: (45584, 22)
```

```
num_fill_cols = ["Delivery_person_Age", "Delivery_person_Ratings", "multiple_deliveries", "prep_time_min"]
cat_fill_cols = ["Weather_conditions", "Road_traffic_density", "Type_of_order",
                 "Type_of_vehicle", "Festival", "City"]

for col in num_fill_cols:
    if col in df.columns:
        df[col] = df[col].fillna(df[col].median())

for col in cat_fill_cols:
    if col in df.columns:
        df[col] = df[col].fillna(df[col].mode()[0])

print(df.isnull().sum())
```

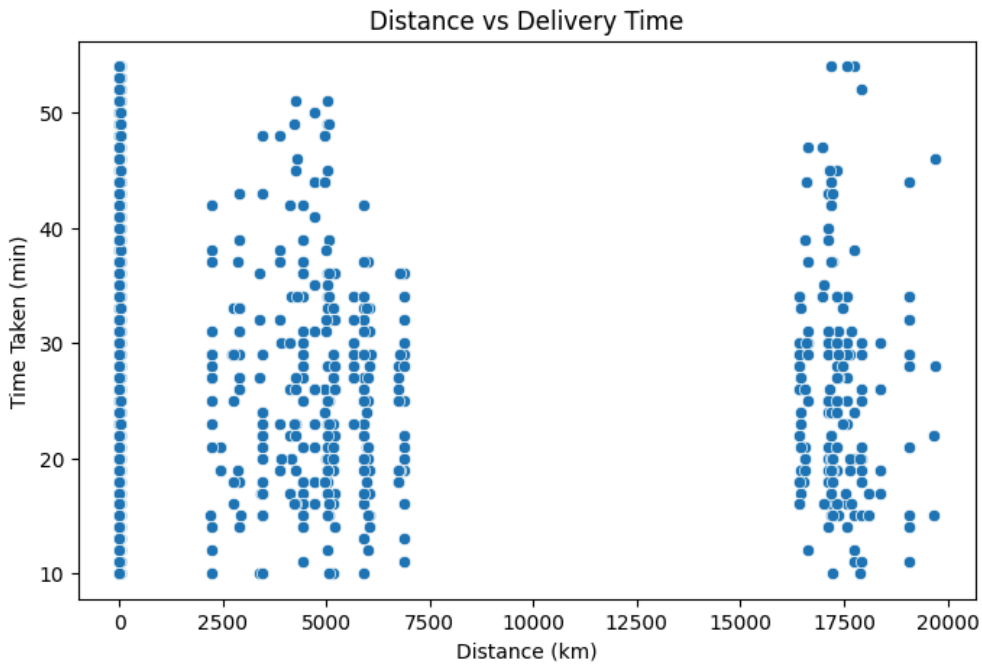
```
ID                             0
Delivery_person_ID             0
Delivery_person_Age            0
Delivery_person_Ratings        0
Restaurant_latitude            0
Restaurant_longitude           0
Delivery_location_latitude     0
Delivery_location_longitude    0
Order_Date                     0
Time_Orderd                 5799
```

```
Time_Order_picked              5007
Weather_conditions                0
Road_traffic_density              0
Vehicle_condition                 0
Type_of_order                     0
Type_of_vehicle                   0
multiple_deliveries               0
Festival                          0
City                              0
Time_taken_min                    0
prep_time_min                     0
distance_km                       0
dtype: int64
```

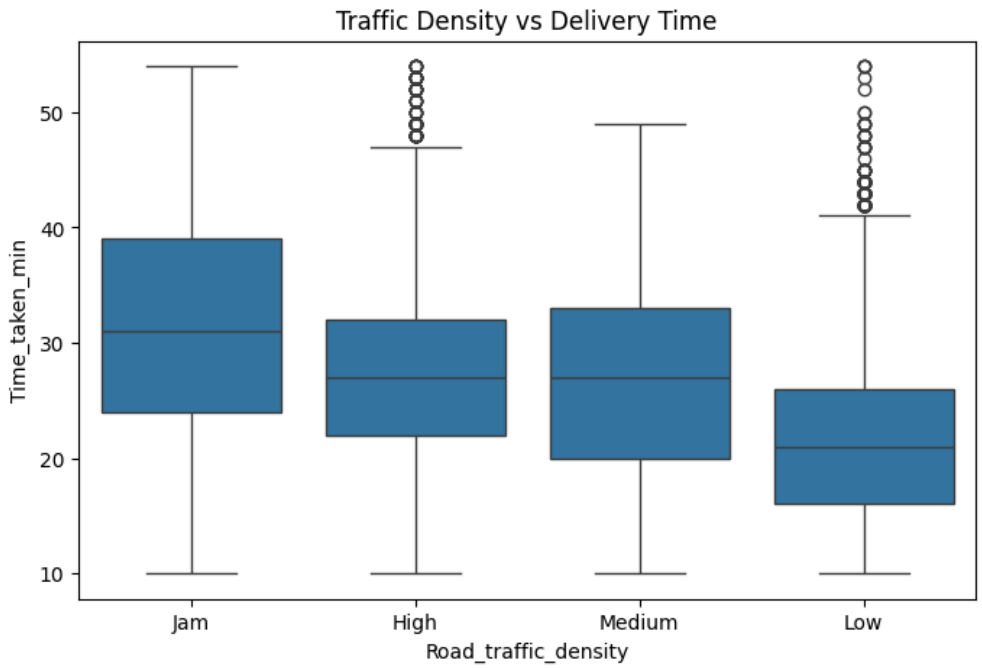
```
plt.figure(figsize=(8,5))
sns.histplot(df["Time_taken_min"], bins=30, kde=True)
plt.title("Distribution of Delivery Time")
plt.xlabel("Time Taken (min)")
plt.ylabel("Frequency")
plt.show()
```




```
plt.figure(figsize=(8,5))
sns.scatterplot(x=df["distance_km"], y=df["Time_taken_min"])
plt.title("Distance vs Delivery Time")
plt.xlabel("Distance (km)")
plt.ylabel("Time Taken (min)")
plt.show()
```

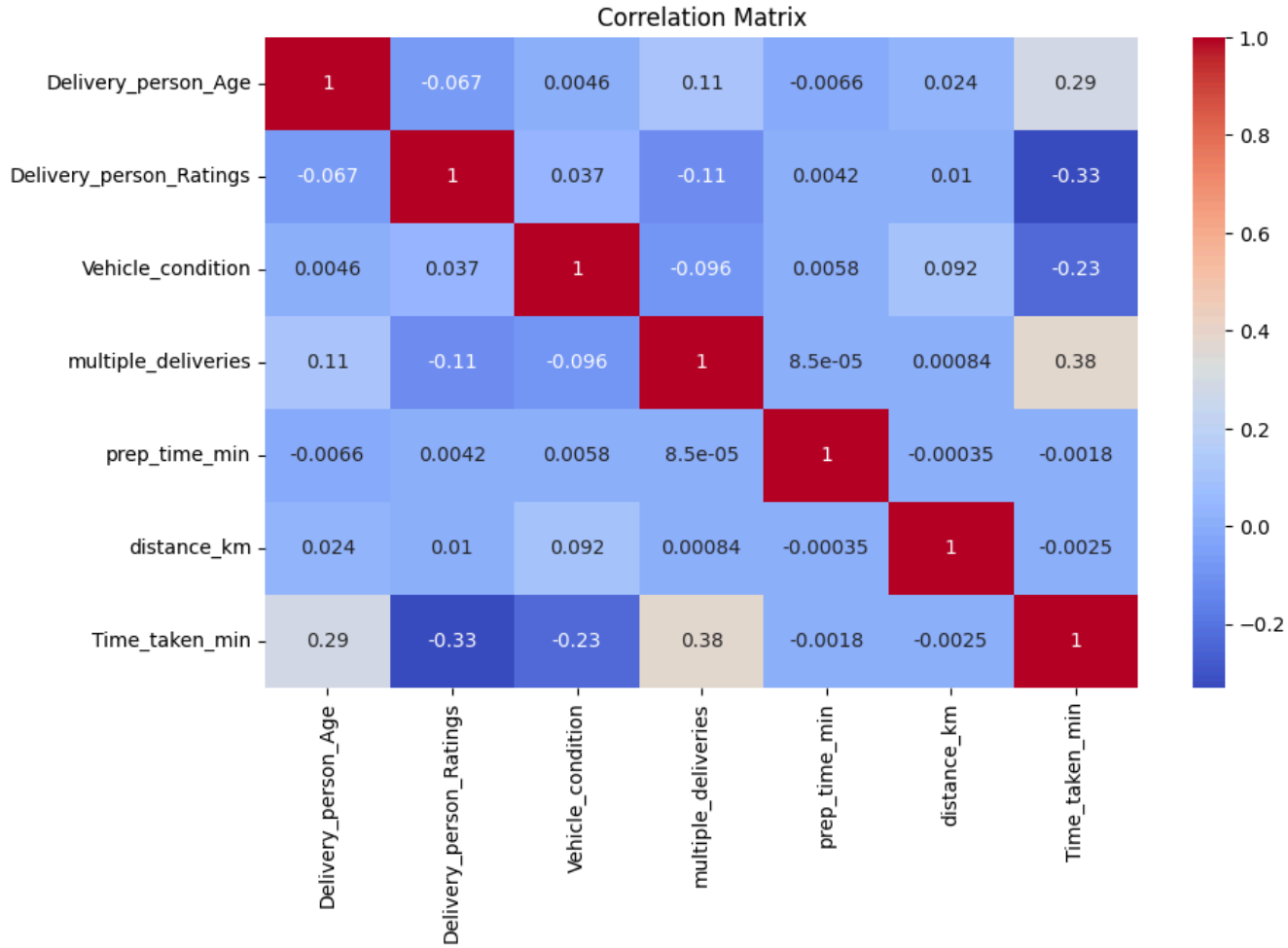


```
plt.figure(figsize=(8,5))
sns.boxplot(x="Road_traffic_density", y="Time_taken_min", data=df)
plt.title("Traffic Density vs Delivery Time")
plt.show()
```




```
corr_cols = ["Delivery_person_Age", "Delivery_person_Ratings", "Vehicle_condition",
             "multiple_deliveries", "prep_time_min", "distance_km", "Time_taken_min"]
```

```
plt.figure(figsize=(10,6))
sns.heatmap(df[corr_cols].corr(), annot=True, cmap="coolwarm")
plt.title("Correlation Matrix")
plt.show()
```




```
from sklearn.model_selection import train_test_split
from sklearn.compose import ColumnTransformer
from sklearn.pipeline import Pipeline
from sklearn.impute import SimpleImputer
```

```python
from sklearn.preprocessing import OneHotEncoder
from sklearn.ensemble import RandomForestRegressor
from sklearn.metrics import mean_absolute_error, mean_squared_error, r2_score

features = [
    "Delivery_person_Age",
    "Delivery_person_Ratings",
    "Vehicle_condition",
    "multiple_deliveries",
    "prep_time_min",
    "distance_km",
    "Weather_conditions",
    "Road_traffic_density",
    "Type_of_order",
    "Type_of_vehicle",
    "Festival",
    "City"
]

target = "Time_taken_min"

X = df[features]
y = df[target]

numeric_features = [
    "Delivery_person_Age",
    "Delivery_person_Ratings",
    "Vehicle_condition",
    "multiple_deliveries",
    "prep_time_min",
    "distance_km"
]

categorical_features = [
    "Weather_conditions",
    "Road_traffic_density",
    "Type_of_order",
    "Type_of_vehicle",
    "Festival",
    "City"
]

numeric_transformer = Pipeline(steps=[
    ("imputer", SimpleImputer(strategy="median"))
])

categorical_transformer = Pipeline(steps=[
    ("imputer", SimpleImputer(strategy="most_frequent")),
    ("onehot", OneHotEncoder(handle_unknown="ignore"))
])

preprocessor = ColumnTransformer(transformers=[
    ("num", numeric_transformer, numeric_features),
    ("cat", categorical_transformer, categorical_features)
])

model = Pipeline(steps=[
    ("preprocessor", preprocessor),
    ("regressor", RandomForestRegressor(n_estimators=100, random_state=42))
])

X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)

model.fit(X_train, y_train)
y_pred = model.predict(X_test)

print("MAE:", mean_absolute_error(y_test, y_pred))
print("RMSE:", np.sqrt(mean_squared_error(y_test, y_pred)))
print("R2 Score:", r2_score(y_test, y_pred))
```

```
MAE: 3.2123494570582425
RMSE: 4.081153981883747
R2 Score: 0.8111812247911403
```

```python
plt.figure(figsize=(7,5))
plt.scatter(y_test, y_pred, alpha=0.6)
plt.xlabel("Actual Delivery Time")
plt.ylabel("Predicted Delivery Time")
plt.title("Actual vs Predicted Delivery Time")
plt.show()
```

```
df.to_csv("cleaned_zomato_dataset.csv", index=False)
print("Cleaned dataset saved successfully.")
```

```
Cleaned dataset saved successfully.
```